https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	t is the independent variable and *s* is the dependent variable **or** vary *t* and measure *s*	**1**
	keep *k* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • spring connected to magnet • vertical rule parallel to spring to determine *s* • rule held in position by a stand • stand resting on the bench • rule labelled and at least one other label from stand, clamp, card, (magnetic) sheet, (cylindrical) magnet, spring	**1**
	s = (new) length/position of spring – original length/position of spring	**1**
	use a micrometer to measure *t*	**1**
	measure *B* using a (calibrated) Hall probe **and** rotate probe until maximum value **or** measure *B* using Hall probe first in one direction, then in the opposite direction and average	**1**

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Question	Answer	Marks
1	**Method of Analysis**	
	plot a graph of s against $\frac{1}{t}$ or equivalent (allow lg s against lg t)	**1**
	relationship valid <u>if</u> a straight line that passes through the origin is produced (for lg s against lg t: relationship valid <u>if</u> a straight line with gradient -1)	**1**
	$Z = \frac{k}{ALB} \times$ gradient (for lg s against lg t: $Z = \frac{k \times 10^{y\text{-intercept}}}{ALB}$)	**1**